# A Hybrid SMILES-DeepSMILES Transformer Approach Improving Fragment Prediction in Fragment-Based Drug Discovery with DeepBERTa

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## Background

**FBDD:** Uses small fragments (<300 Da) as drug starting points.

### Problem

- **SMILES:** Paired rings, nested brackets → parsing errors
- Slow processing, invalid outputs

### Solution

**DeepSMILES:** Postfix variant, simpler syntax

**DeepBERTa:** ChemBERTa on DeepSMILES

**Hybrid:** Best of both models

## Methods

1. ~34k FDA drugs, RECAP fragmentation
2. Target: mTOR kinase
3. AutoDock Vina docking
4. Pretrain: 100k ZINC molecules
5. RoBERTa-base architecture
6. Fine-tune with validity penalties
7. Ensemble best predictions

## Key Insights

- DeepSMILES wins **13%** of cases
- SMILES wins **26%** of cases
- Hybrid beats both individually
- **30%** high similarity (>0.8)
- Pretraining scale critical (100k+ needed)

### Limitations

- Single target (mTOR)
- DeepSMILES: 31% vs SMILES: 40% validity
- 34k pairs may limit generalization

## Future Work

- **Structure-based:** Add protein binding site data
- **RL optimization:** Iterative fragment growing
- **Multi-target:** Test beyond mTOR
- **Ensemble expansion:** Add SELFIES, other notations
- **Data augmentation:** Larger, diverse datasets

### Conclusion

**DeepBERTa** + hybrid ensemble outperforms individual models. Validates ensemble molecular representations for cheminformatics.

## Results

**Hybrid approach: 19.7% improvement over SMILES alone (0.290 → 0.347 mean Tanimoto similarity)**

| Metric | SMILES | DeepSMILES | Hybrid Combined |
|---|---|---|---|
| **Mean Tanimoto Similarity** | 0.29 | 0.18 | **0.34** |
| **Chemical Validity (%)** | 42.0 | 31.0 | **52.0** |
| **Tanimoto (Valid Only)** | 0.40 | 0.31 | **0.47** |
| **Predictions > 0.8 Tanimoto (%)** | 26.4 | 18.9 | **29.7** |
| **Sequence Matcher Ratio** | 0.74 | 0.67 | 0.74 |